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Electrical characteristics of ore-forming intrusion No. 2 in the Xiarihamu magmatic Ni-Cu sulfide deposit in East Kunlun Orogenic Belt

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ABSTRACT

The Xiarihamu magmatic Ni-Cu sulfide deposit is situated within the East Kunlun Orogenic Belt and consists of four intrusions. While the estimated Ni resources of intrusion No. 1 exceed 1.2 million tons, the resource potential of other intrusions remains unknown. One significant factor contributing to this knowledge gap is the lack of effective detection and localization methods for concealed intrusions. To address this issue, we conducted an electromagnetic study focusing on intrusion No. 2. Due to the substantial topographic relief in the survey area, the conventional artificial-source electromagnetic method presented limitations in data acquisition. Consequently, we employed the short-offset transient electromagnetic method for data acquisition. A group-network observation mode was utilized to reconstruct the electrical structure of intrusion No. 2 through data inversion. Our results reveal a predominant high-resistivity background within the survey area. Rocks including granite, biotite gneiss, gabbro, and pyroxenite exhibit high resistivities, whereas mineralized ultramafic rocks show lower resistivities due to the presence of sulfides. The detection of concealed conductive bodies serves as a crucial indicator for deep mineral exploration. The distribution characteristics of low-resistivity anomalies indicate that the mafic bodies extend deeper towards the southwest, with mineralized pyroxenites observed beneath the gabbros. The results also suggest a gradual weakening of mineralization from the northeast to the southwest, with conductive mineralized anomalies existing at considerable depths beneath the known exposed intrusion. These findings highlight promising prospecting potential and provide valuable insights into the formation of intrusions within the Xiarihamu deposit, supporting resource exploration in similar deposits.

1. Introduction

Magmatic nickel-copper (Ni-Cu) sulfide deposits typically form within stable cratons (Naldrett, 2004; Queffurus and Barnes, 2014; Barnes et al., 2016; Yuan and Su, 2023). However, in China, such deposits also occur within orogenic belts (Mao et al., 2018; Wei et al., 2019; Tang et al., 2020; Cui et al., 2022). The Xiarihamu magmatic Ni-Cu sulfide deposit is the first discovery of such deposits within the East Kunlun Orogenic Belt (EKOB) (Song et al., 2016; Zhang et al., 2017; Liu et al., 2018; Li et al., 2023). Fig. 1 illustrates the location of the study area that is located in the northern Qinghai–Tibet Plateau.

Due to the presence of mafic–ultramafic intrusion in the Xiarihamu deposit, direct detection of the ore body poses challenges. Prospecting efforts for ore bodies should commence by searching for ultramafic intrusions. Previous geological investigations have identified four main ore-forming intrusions. Intrusion No. 1 is primarily composed of pyroxenites and has a confirmed Ni resource exceeding 1.2 million tons (Zhang et al., 2017). The other exposed intrusions are predominantly composed of gabbro and do not contain economically viable ore bodies on the surface (Bao et al., 2023). For instance, in the case of intrusion No. 2, the exposed mafic intrusion at the surface is limited, and the main challenges in locating the ore body within this intrusion lie in determining deep extension of the mafic body and occurrence of ultramafic body.

Geophysical methods play a crucial role in the exploration of magmatic-type Ni-Cu deposits (Su et al., 2023a, b; Xue et al., 2024), with the transient electromagnetic (TEM) method being particularly significant. The TEM method has proven effective in the deep localization of ore bodies across various types of mineral deposits (Di et al., 2018; Guo et al., 2020; Xue et al., 2024). In general, mafic–ultramafic intrusions

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exhibit higher resistivities compared to the surrounding rocks, whereas mineralized or altered bodies typically display relatively lower resistivities (Zhou et al., 2022). For the electromagnetic exploration of the Xiarihamu deposit, accurate identification of resistive non-mineralized and conductive mineralized rock bodies is a critical challenge (Xue et al., 2023). The survey area for this study is situated in a branch of the East Kunlun Orogenic Belt, characterized by rugged terrain with deep cuts and steep mountain peaks, ranging in altitude from 3,200 to 3,600 m. Consequently, it is imperative to employ appropriate TEM configurations for conducting field surveys in such complex terrains.

The TEM configurations primarily fall into two types. The first type involves a loop source that allows for the observation of the electromagnetic field either inside or outside the transmitting loop. However, deploying the loop source is challenging and not suitable for mountainous areas (Xue et al., 2007a, b, 2012). The second configuration utilizes a grounded wire source, which adopts a separate observation mode for the source and measurement points, with a certain distance between transmitting source and measurement line. Importantly, this source is easier to deploy. Previous studies have employed portable receiving devices that are tailored to the terrain characteristics of deep-cut areas in mountainous regions to observe the electromagnetic fields (Hordt et al., 2000; Wright, 2003; Haber et al., 2004; Di, 2015; Di et al., 2018). The TEM configuration with a grounded wire source is also referred to as the electrical-source TEM configuration. There are two main observation modes based on the ratio between the transmitter–receiver distance and the target depth: the short-offset transient electromagnetic (SOTEM) mode and the long-offset (LOTEM) mode. The SOTEM, where the ratio is smaller than 2, offers stronger detectability compared to the long-offset mode. As a result, it is the most commonly used working device for the electrical-source TEM configuration and plays a vital role in mineral deposit exploration (Xue et al., 2014).

In this study, our initial focus was on analyzing the resistivity characteristics of the main rocks within the Xiarihamu deposit. Subsequently, we proceeded with an exploration using the SOTEM method specifically targeting intrusion No. 2. The objective was to ascertain the electrical characteristics of the intrusion and accurately locate the mineralized body within the deposit.

2. Geological background of the Xiarihamu deposit

The Xiarihamu deposit originated during the tectonic regime stage of the Middle and Late Silurian to Early/Middle Devonian period (Bao et al., 2023; Li et al., 2023). It underwent a transformation from a syn-collisional extrusion to a post-collisional extension setting. The Xiarihamu survey area encompasses four primary mafic–ultramafic bodies. Intrusion Nos. 2 and 4 consist primarily of gabbros, which are exposed at the surface, and intrusion Nos. 1 and 3 predominantly consist of pyroxenites (Fig. 2). The known ore deposits are primarily hosted within the ultramafic bodies. Regarding intrusion Nos. 2 and 4, no ore bodies have been identified within the mafic bodies at the surface. Ultramafic bodies may exist at deeper levels within these intrusions, but their precise locations remain unknown at present.

3. Methods and data

3.1. Methods

The electrical-source TEM method utilizes a distinct observation setup, wherein a grounded wire serves as the transmitting source, and the receiving line is positioned at a specific offset from the transmitting source (Xue et al., 2014). Within this method, the short-offset transient electromagnetic (SOTEM) configuration has been recently proposed. It deviates from the conventionally employed long-offset mode, which is predominantly employed for oil and gas exploration purposes. The SOTEM involves a near-source observation approach that yields robust signal strengths and precise detection capabilities for geoelectric targets. Consequently, it is better suited for mineral resource exploration.

The horizontal and vertical magnetic fields were the two primary components that were observed. The observation of the electric field required strict adherence to the designated measurement point locations. However, due to the complex terrain of the survey area, it was challenging to accurately position the grounding electrodes at the allocated measurement points. Consequently, obtaining precise electric field measurements proved difficult. However, the magnetic field was observed using a flexible probe that could be laid out more easily. This allowed for accurate observation of the vertical magnetic field component. Considering the complexity of the terrain, a group-network observation mode was implemented during the field survey. Data were collected within a certain area, and measurement points were selected to form survey lines, as opposed to pre-designing the survey lines before data collection.

The expression for the vertical magnetic field component used in this study was derived from the work of Ward and Hohmann (1991). The specific expression is as follows:

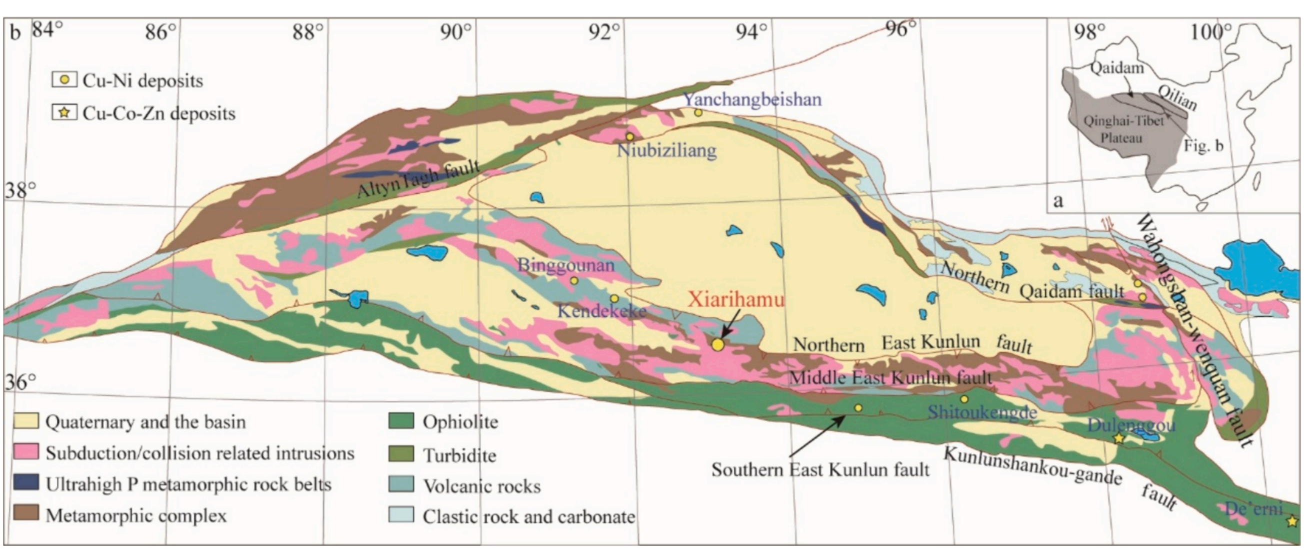

Fig. 1. (a) The location of the Qaidam block in the northeastern region of the Qinghai-Tibet Plateau. (b) A geological sketch map depicting the location of the Xiarihamu deposit in the East Kunlun Orogenic Belt (modified after Tang et al., 2024).

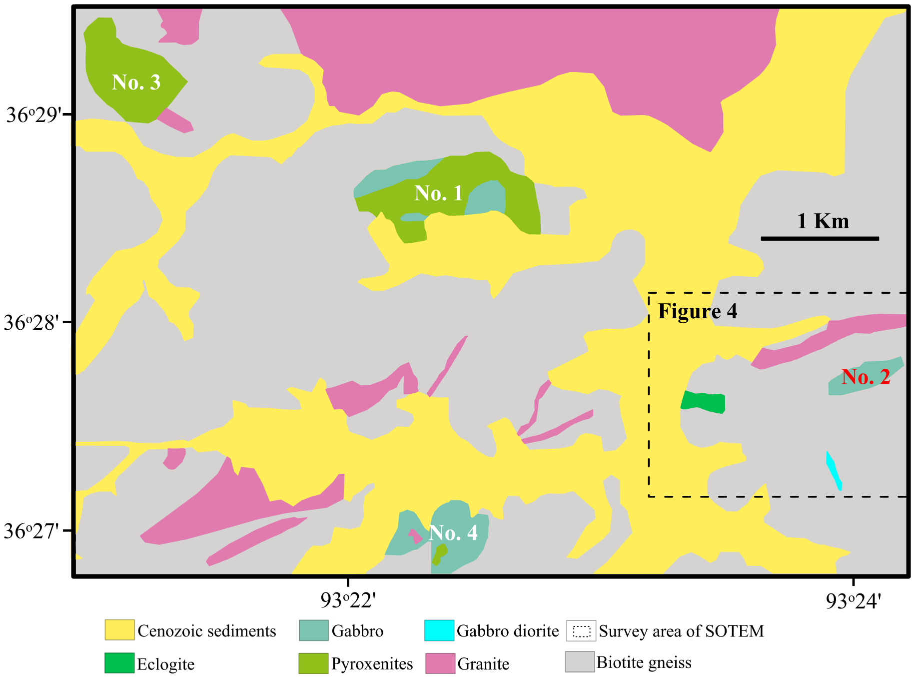

Fig. 2. Simplified geological map of the four intrusions in the Xiarihamu magmatic Ni-Cu sulfide deposit (modified after Zhang et al., 2017). The black dashed line indicates the survey area of the short-offset transient electromagnetic (SOTEM) method conducted on intrusion No. 2.

$$H_z = \frac{I}{4\pi}\int_{-L}^{L}\frac{y}{r}\int_0^{\infty}(1+r_{TE})e^{u_0 z}\frac{\lambda^2}{u_0}J_1(\lambda r)d\lambda dx', \tag{1}$$

where λ is the wave number; $J_1(\lambda r)$ is the Bessel function of the first kind; L is half length of the source; and r_{TE} is the reflection coefficient related to the transverse electric field.

Based on the available geological information of the survey area, it is known that the orebody is primarily hosted in pyroxenites, which exhibit a relatively low resistivity. In contrast, non-mineralized gabbros and ultramafic pyroxenites have high resistivities. This contrast in resistivity serves as the foundation for conducting the SOTEM survey, as shown in Table 1.

To assess the feasibility of using the vertical magnetic field in detecting mineralized bodies with low resistivities, a forward simulation was conducted. The electrical model employed a three-layer design. The mineralized body was characterized by a low resistivity of 400 Ωm in comparison to the resistivity of the homogeneous earth (2,000 Ωm). The thickness of the mineralized bodies varies from 10 m to 100 m. The response difference resulting from the low-resistivity anomaly was analyzed.

We observed that low-resistivity mineralized intrusions exhibit significant response differences compared to the uniform earth (Fig. 3, black line). As the thickness of the low-resistivity body increased, the decay in response was more pronounced during the early period until 1E-5 s, but show a stronger response thereafter, from 1E-4 s onwards. This indicates an enhanced response strength and a greater contrast between the low-resistivity mineralized intrusion and the uniform earth.

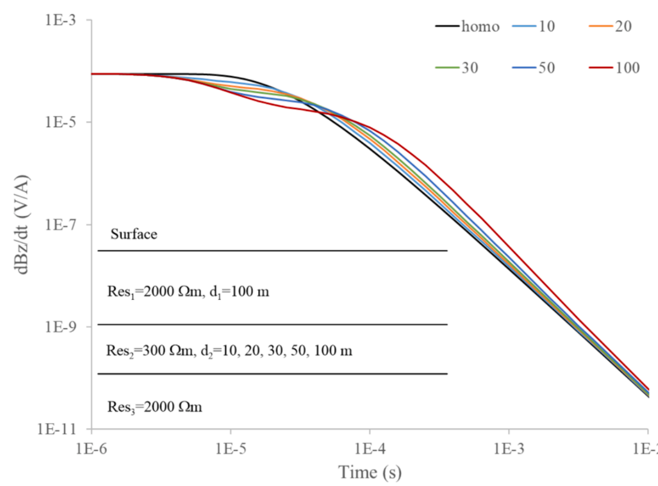

Fig. 3. Response characteristics of mineralized bodies with varying thicknesses and low resistivities. The colored lines represent the response corresponding to bodies with different thicknesses.

These findings suggest the feasibility of using the vertical magnetic field for detecting low-resistivity mineralized intrusions.

Table 1
Resistivity of rocks in the Xiarihamu deposit.

Rock/ore	Number of samples	Resistivity (Ωm)
Gabbro	40	2,021
Pyroxenite	24	3,072
Mineralized pyroxenite	32	330
Biotite gneiss	24	1,809
Granite	11	2,432

3.2. Data acquisition and analysis

Due to the undulating terrain, the survey approach employed in this study involved initial area observations, followed by the selection of measurement lines based on detection requirements. The specific locations of transmitting source deployment and measurement points can be observed in Fig. 4. Based on our geological investigations, the exposed gabbro body at the surface likely extends towards the southwest at greater depths. Consequently, we chose the measurement line indicated by the solid blue line to align with this potential extension. In terms of the observation parameters, the base frequencies for our measurements were 25 Hz and 5 Hz, and the time range considered was from 0.079 ms

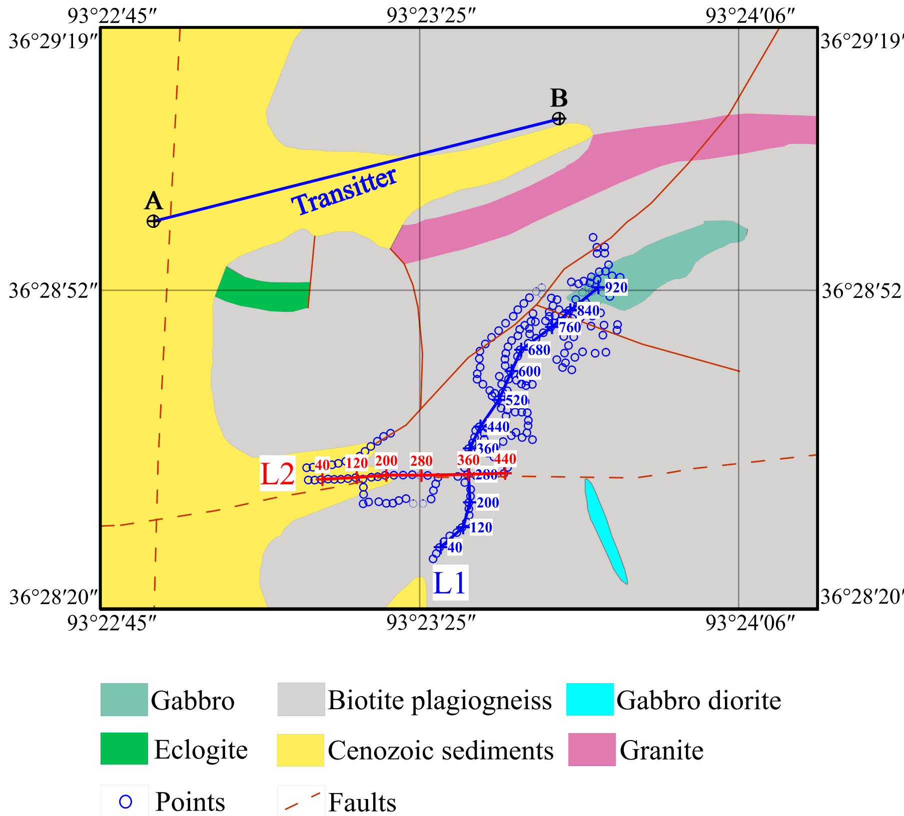

Fig. 4. Geological map of intrusion No. 2 along the layout of survey lines and the transmitter used in the SOTEM survey conducted in this study. The transmitter is denoted by the label A-B, and the measurement points are represented by blue circles. The survey lines, depicted by blue and red lines, are labeled as L1 and L2, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to 42.61 ms. It is important to note that the early-time responses, which did not exhibit the expected attenuation characteristics, were primarily influenced by the near-zone effect. As a result, these responses were truncated before further data processing. For the analysis and inversion processes, we specifically selected the monotonic attenuation response for further examination.

In Fig. 5, the response curves measured at 400 m and 900 m measurement points along line L1 are presented. Notably, the response curve at the 900 m point exhibits a faster decay rate before 3 ms compared to the response curve at the 400 m point. The response strength at the 900 m point is weaker than that at the 400 m during this early period. However, after 5 ms, the response at the 900 m point decays more slowly, and its strength is higher than that at the 400 m. Based on the decay pattern observed in the response curves and considering the findings from the forward simulation (Fig. 3), we suggest the presence of a low-resistivity anomaly beneath the 900 m point.

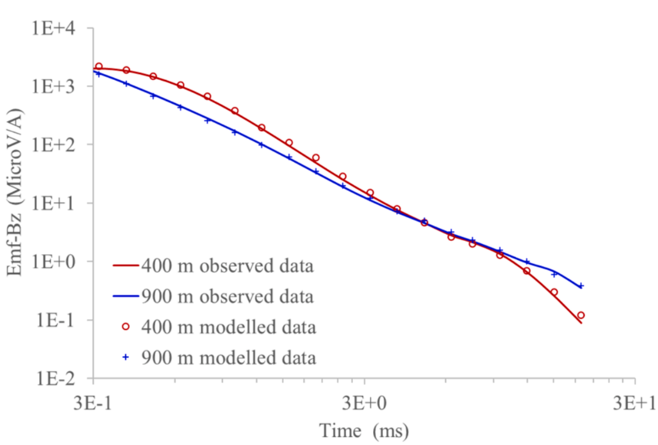

Fig. 5. Observed and modeled response curves at two specific distances along line L1: 400 m and 900 m. The response curves at 400 m are depicted in red, while the response curves at 900 m are in blue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

We employed the Occam algorithm, a commonly used inversion method for TEM data, to invert the measured data as described in the study by Li et al. (2019). The observed data and modeled data are in good agreement, as shown in Fig. 5. The results of the data inversion are presented in Fig. 6. Based on the inversion results, we identified a low-resistivity body with a resistivity lower than 400 Ωm in the depth range of 50–150 m at the 900 m measurement point. This low-resistivity body is inferred to be a mineralized ultramafic intrusion. In contrast, at the 400 m measurement point, the resistivity in the corresponding depth range is high, exceeding 1,000 Ωm. Furthermore, the inversion results for both measurement points at a depth of 200 m depicted relatively low resistivities. This suggests the presence of a weakly mineralized ultramafic body at that depth.

4. Results and discussion

In Fig. 7, the shallow portion ranging from 700 to 920 m is depicted in yellow color, indicating resistivity values exceeding 2,500 Ωm. This yellow region corresponds to an exposed gabbro body, as determined through the geological investigation conducted in this study. A distinct blue low-resistivity body is observed within the depth range of 80 to 150 m. This blue region is inferred to be mineralized pyroxenite body. Notably, as the measurement point progresses towards the southwestern direction, the depth of the blue low-resistivity body increases. This is evident as the blue color shifts from dark to light blue until the 600 m point along the survey line. Deeper areas where extensions of yellow-

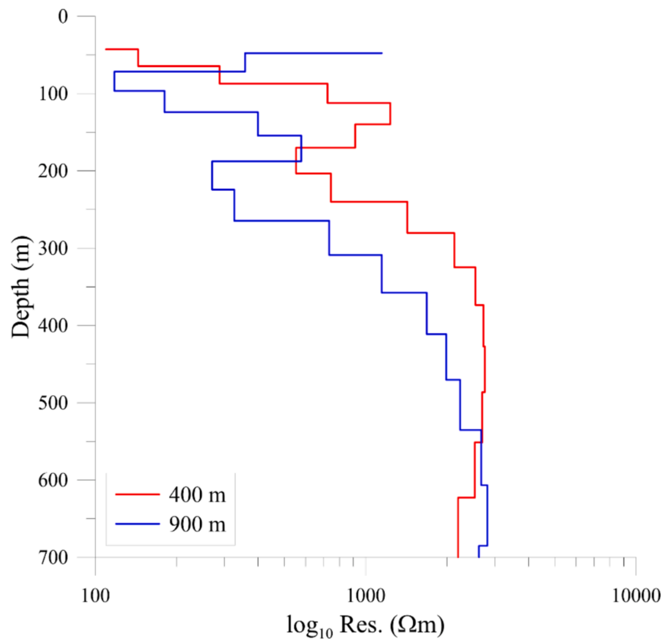

Fig. 6. Inverted resistivity results for the measurement points located at 400 m (red line) and 900 m (blue line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

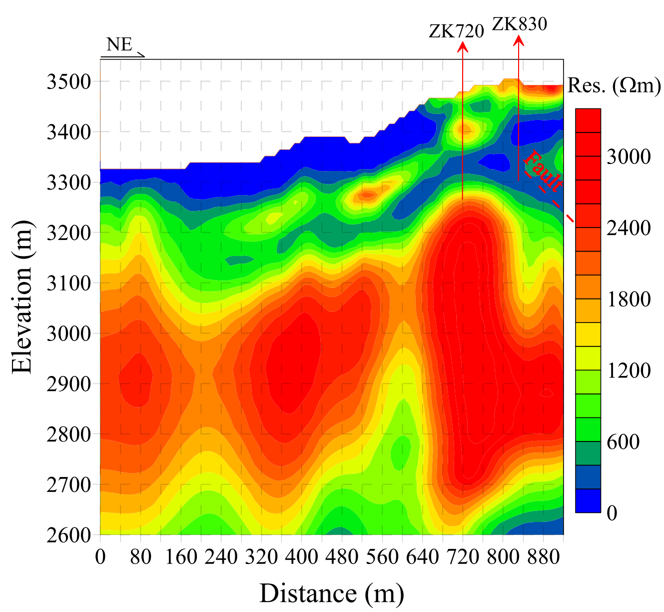

Fig. 7. Inverted resistivity plotted with respect to elevation along survey line L1. High resistivity values are depicted in red, and low resistivity values are in blue. The drilling points along the survey line are denoted by ZK. A fault is indicated by a red dashed line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

orange and green colors are observed can be identified as gabbro and weakly mineralized pyroxenite bodies, respectively. Furthermore, the inferred fault, as represented by the red dashed line, aligns with the fault observed in the geological map shown in Fig. 4.

Along survey line L2, as shown in Fig. 8, similar electrical characteristics were observed. Specifically, within the elevation range of 3,200–3,100 m, a relatively low-resistivity anomaly was identified between 100 and 300 m. This low-resistivity anomaly is located below a high-resistivity anomaly, which is inferred to represent a gabbro body. Below the high-resistivity gabbro body, there is a low-resistivity region,

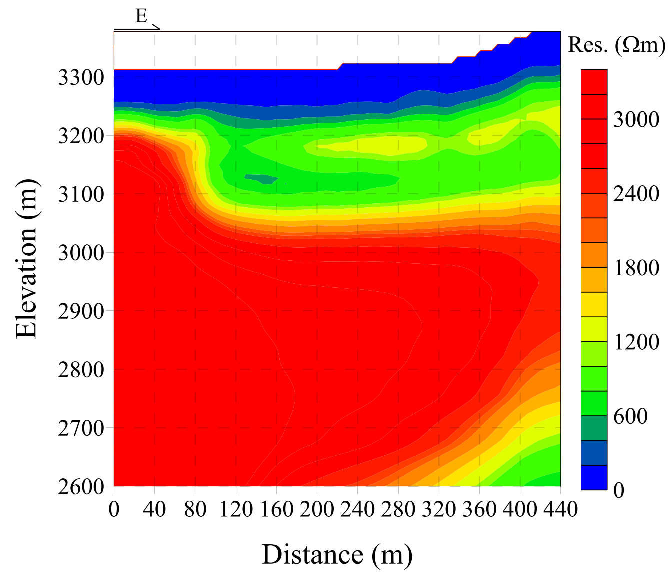

Fig. 8. Inverted resistivity plotted with respect to elevation along survey line L2. High resistivity values are represented in red, while low resistivity values are in blue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

indicating the presence of a weakly mineralized pyroxenite body.

Fig. 9 represents a three-dimensional display of both survey lines L1 and L2. The visualization reveals that the two low-resistivity anomalies observed along the survey lines are spatially coherent with each other, suggesting that they are part of the same mineralized body rather than separate entities. The inversion results for survey lines L1 and L2 (Fig. 9) helped delineate the stratigraphic structure at different depths. From shallow to deep depths, the identified stratigraphic units include Cenozoic sediments, biotite gneiss, gabbro, mineralized pyroxenite, granite, and granite gneiss.

The presence of gabbro was observed in the shallow part at measuring points beyond 700 m, while mineralized pyroxenite was found in the lower part or at the bottom of the gabbro. These findings were verified by drilling results. Two boreholes were drilled along survey line L1, specifically at the 720 m and 830 m points (as depicted in Fig. 7). At the 720 m point, an ore body with a thickness of 10.95 m was discovered in the elevation range of 3347–3333 m, containing Ni, Cu, and Co with grades of 0.94 %, 0.31 %, and 0.03 %, respectively. Similarly, at the 830 m point, an ore body with a thickness of 47.76 m was found in the elevation range of 3428–3363 m, with Ni, Cu, and Co grades of 0.94 %, 0.31 %, and 0.03 %, respectively. There is a correlation between the thickness of the ore body and the thickness of the blue low-resistivity body, indicating a positive relationship. Thicker ore bodies exhibit thicker blue low-resistivity anomalies.

Overall, the conductive anomalies decrease from northeast to southwest, and the color of the low resistivity gradually changes from blue to green, suggesting a gradual decrease in the thickness of the mineralized pyroxenite and the corresponding ore body. However, mineralized pyroxenites are still present in the deep southwest region, indicating a potential target for future mineral exploration. The possibility of deep mineral deposits beneath the exposed mineralized bodies in the northeast direction remains, highlighting it as an important area for future research.

Based on the spatial relationship observed between the mineralized pyroxenite body and faults in the region, it is inferred that deep magma upwelled along the faults into the shallow part of the subsurface. Once reaching the shallow gneiss strata, the magma migrated in two directions and eventually developed at the bottom of the gabbro body. This inference is supported by the fact that the mineralized pyroxenites have a greater density (3.086×10^3 kg/m^3) compared to the gabbro

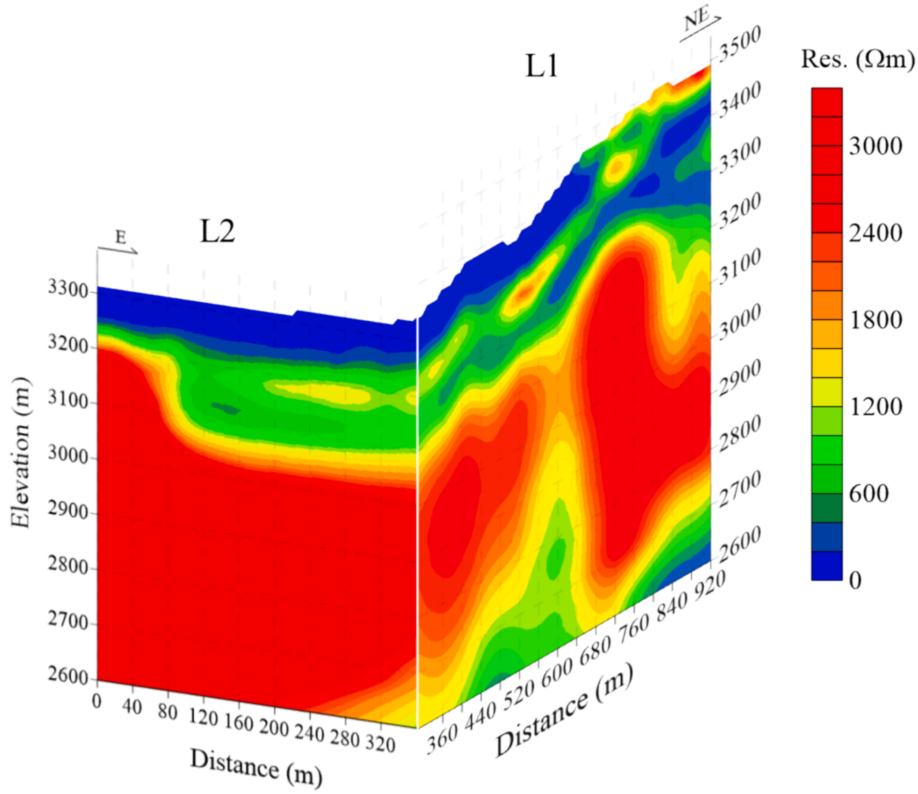

Fig. 9. Stereoscopic slice of an inverted resistivity-elevation section.

(2.878×10^3 kg/m^3) (Fig. 10).

However, this study does have a few limitations. Firstly, due to the influence of complex topography, the electrical characteristics of intrusion No. 2 could not be fully analyzed along the northeast direction. This limitation suggests that there may be additional insights or details about intrusion No. 2 that could not be captured or fully understood in the study. Additionally, the deep fractures in the region that correspond to magma upwelling need to be further characterized. Understanding the nature and characteristics of these deep fractures can provide valuable information about the pathways and mechanisms through which magma migrated and interacted with the surrounding rocks.

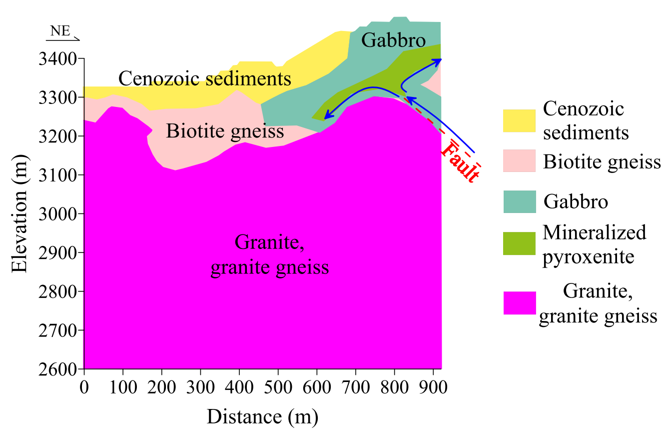

Fig. 10. Geological model of survey line L2. Blue arrows denote the migration direction of magma. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5. Conclusions

In this study, electrical characteristics of intrusion No. 2 of the Xiarihamu deposit were determined using a SOTEM survey. Our results revealed that the exposed gabbro body at the surface extends to greater depths towards the southwest. A low-resistivity mineralized pyroxenite body was identified beneath the mafic intrusion. This hidden conductive body is considered an important indicator for mineral exploration. These results indicate the presence of mineralization, although the degree of mineralization gradually weakens from the northeast to the southwest, and the thickness of the ore body decreases accordingly. However, notable low-resistivity anomalies were observed in the deeper areas associated with known exposed mafic and ultramafic intrusions, suggesting the existence of deep mineralization and good potential for further prospecting.

Based on the inversion results, it was determined that the ultramafic bodies extend deeper toward the southwest, reaching depths of up to 200 m. This suggests that deep magma migrated upward along high-angle fractures and subsequently moved horizontally upon the gneiss strata. In general, mineralized pyroxenites are denser than gabbros, and it is thus inferred that the mineralized pyroxenite may be located in the lower part or at the bottom of the gabbro body. These findings contribute to the understanding of the subsurface geology, mineralization patterns, and potential prospects for further mineral exploration in the study area.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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